Modelling Nonlinear Relationships in Financial Markets: VIX vs High-Yield Spreads

Objective

Nonlinearity is a diagnostic problem, not just a modelling choice

- The goal is to determine whether the mapping is stable or state-dependent, and whether simple linear assumptions are defensible

Workflow framing:

- **Detect** (is there curvature / regime behaviour)
- **Model** (try several functional forms)
- **Diagnose** (residuals, stability, tail behaviour, regime consistency)

Why not "pick the winner by RMSE":

- In finance, the key risk is often **tail / regime error**
- A **model can win on average RMSE** and **still be wrong in stress—the regime you care about most**

Why Care About Detecting Nonlinearity

Risk is state-dependent:

- Relationships often behave differently in calm vs stress
- In calm markets, shocks have roughly proportional effects; in stress, feedback and liquidity constraints can amplify even small shocks, making impacts larger and more persistent (often with lags)

Better stress testing & scenario sizing:

- Nonlinearity changes how shocks propagate (e.g., a 10% move can have a much larger effect in stressed states than in normal states)
- **Contagion/correlation rises**

Avoid false confidence:

- Linear models and ML models trained mostly on **patterns dominated by the calm regime** can look good in-sample yet become **problems during regime shift**
- Detecting nonlinearity is an early warning that you need **regime features**, **robust transforms**, or **segmented models**

Model selection becomes more disciplined:

- Once nonlinearity is established, you can justify transforms, regime conditioning, or monotone/constraint choices **rather than "curve fitting"**

Data Example

VIX

- Market **implied volatility** of the S&P 500 over ~30 days
- Common **risk-off / fear gauge**: higher VIX ⇒ higher perceived equity risk

HY OAS (BAMLH0A0HYM2) - US sub-investment-grade corporate bonds(callable options…)

- **Option-adjusted spread** of a broad US high-yield corporate bond index
- Spread over the **US Treasury curve**, expressed in bps (credit + liquidity premium)
- **Higher HY OAS** - markets are demanding **more compensation for credit risk / liquidity stress** (risk-off)

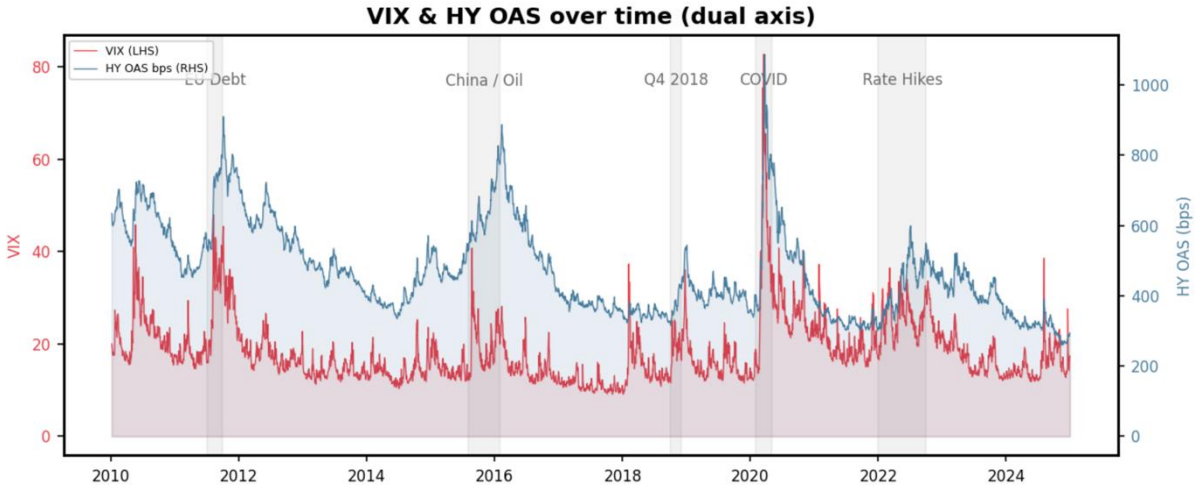
Relationship (why we expect nonlinearity)

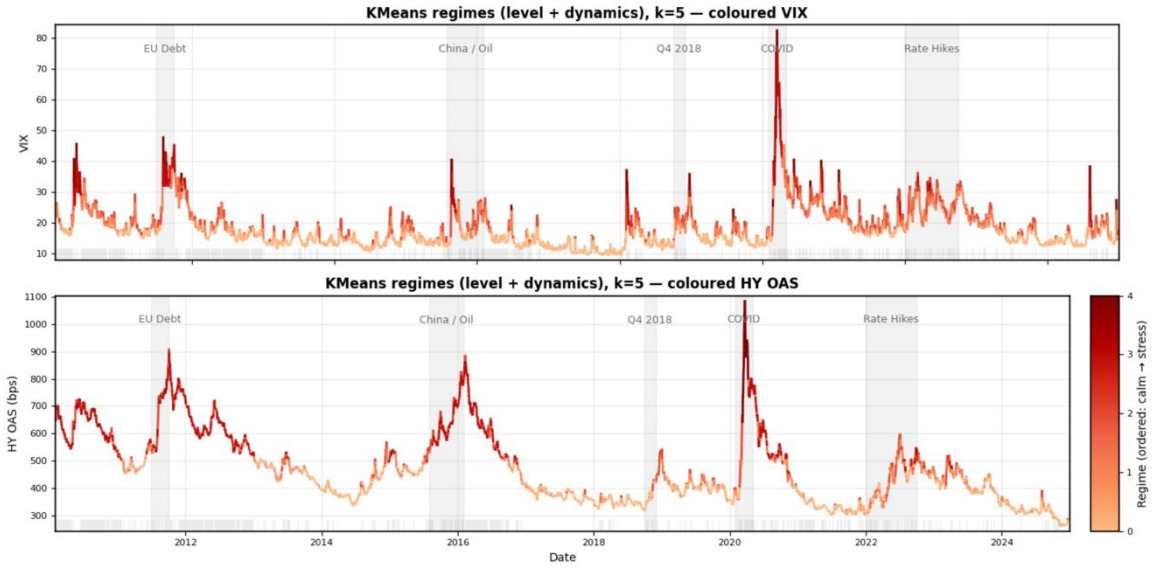
- When **markets de-risk**, implied volatility rises (VIX up), high-yield spreads widen (HY OAS up), and risk assets **sell/risk-off**
- **VIX vs. Spread** linkage is non-linear/regime-dependent - smaller in calm conditions but amplified/convex in stress due to liquidity and risk-aversion feedback

Risk-off is a market regime where investors reduce exposure to risky assets and shift toward safer, more liquid holdings

Data

- **VIX** in points (e.g., 15–30 normally, spikes in crises)
- **HY OAS** (BofA/ICE series) converted from percent to **basis points**





Top Chart

- Shared shock timing, different persistence
- Major episodes (EU debt, China/oil, Q4'18, COVID, rate hikes) show up in **both** series
- VIX reacts fast and mean-reverts quickly
- **OAS is slower and "sticky"** (stress persists after the volatility spike)

Bottom Chart

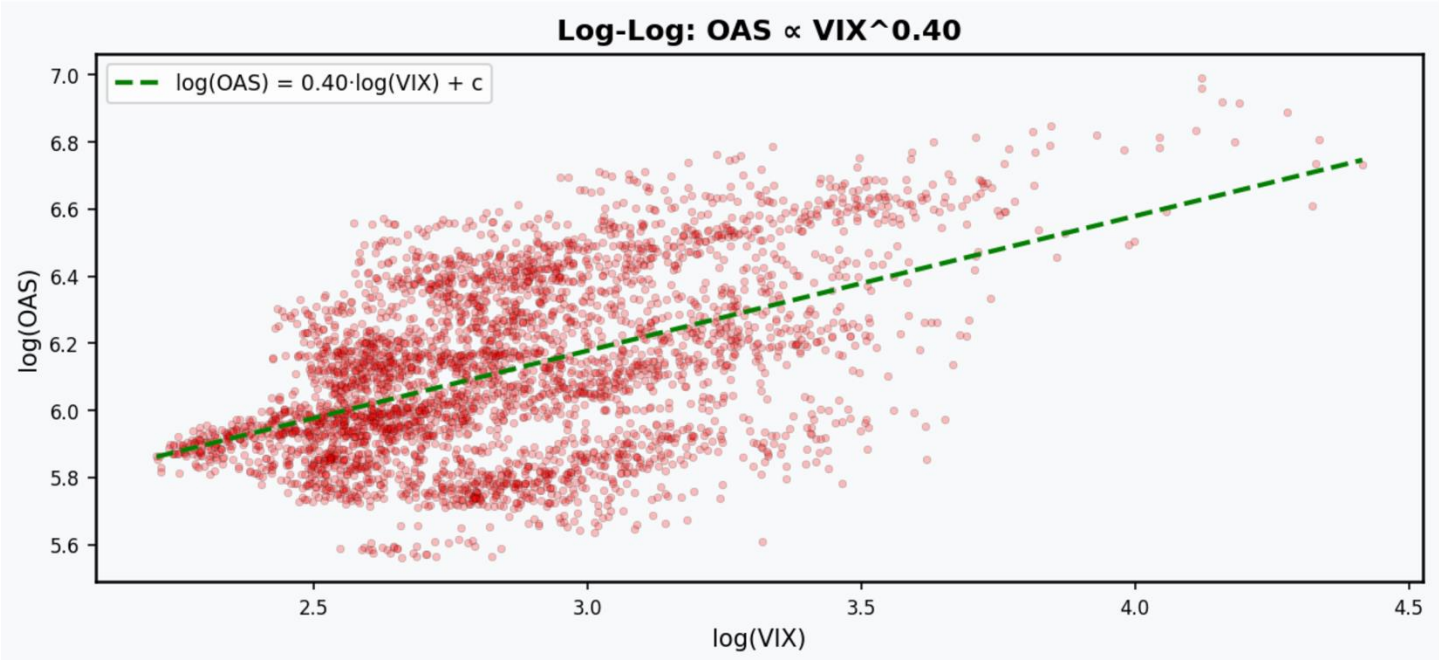
- K-Means - **unsupervised regime detector – does not know time**
- **Regimes are not "true states"**
- KMeans (k=5, dynamics) labels market **states calm / stress**
- Stress regimes align strongly in big crises, but **duration differs**: OAS stays stressed longer than VIX

Relationship is regime-dependent: can't assume one stable linear mapping from VIX to OAS across all market states

Data

Raw levels

- **VIX** in points (e.g., 15–30 normally, spikes in crises).
- **HY OAS** (BofA/ICE series) converted from percent to **basis points**.



The relationship is not well-captured by a single straight line in raw levels

Transformation:

1. log_VIX = log(VIX)
2. log_OAS = log(OAS_bps)

$$\log(OAS) = \alpha + \beta \log(VIX) \Rightarrow OAS \propto VIX^{\beta}$$

Data

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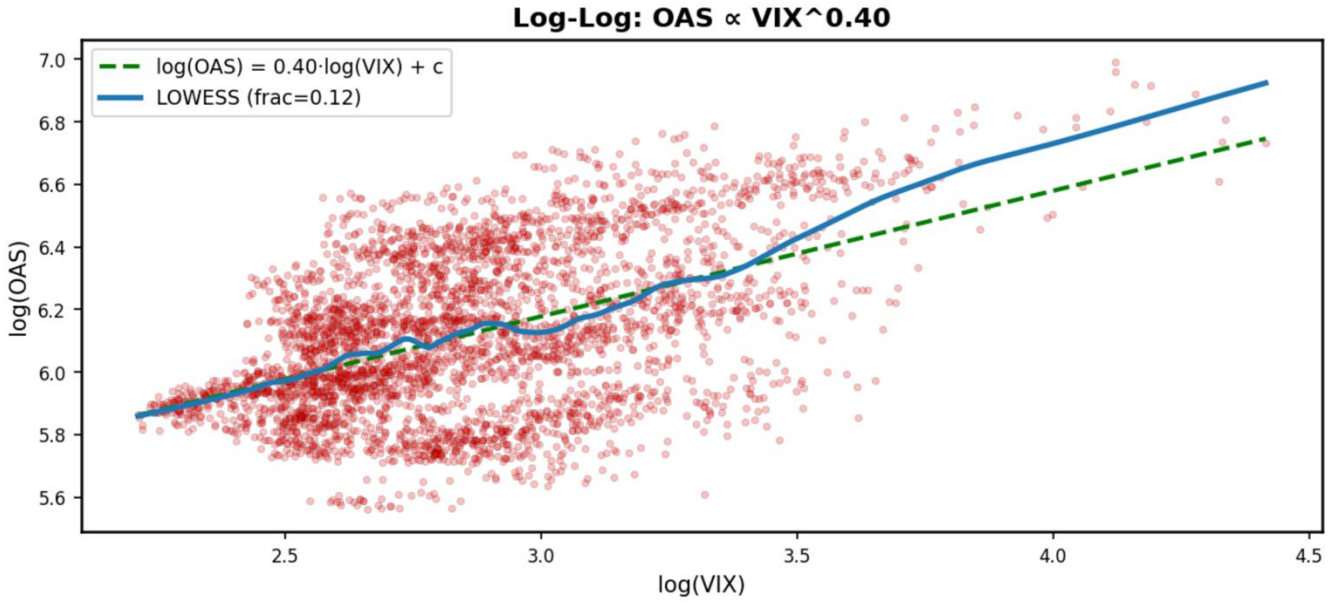
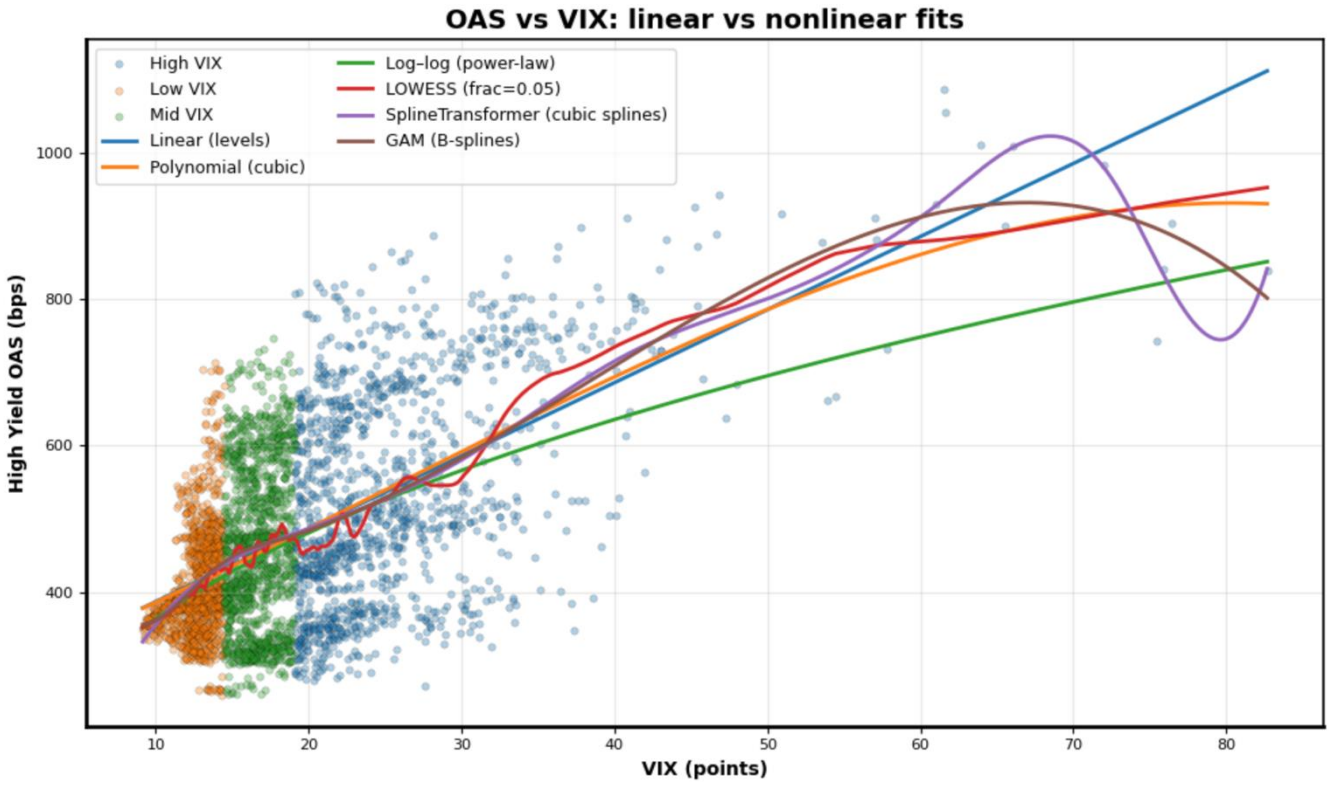


- **Positive relationship, but not perfectly linear in log–log space : S-shape curvature**
- Flatter / wavy in the mid-region (around log(VIX) ≈ 2.8–3.2)
- **Steepens materially at higher log(VIX)** (roughly > 3.3–3.5) - **Implies Convexity**
  - When volatility is already high, additional increases in VIX correspond to **disproportionately**
- **Single beta (0.40) is a good baseline, but the LOWESS suggests you'll improve realism with nonlinear structure**

Models

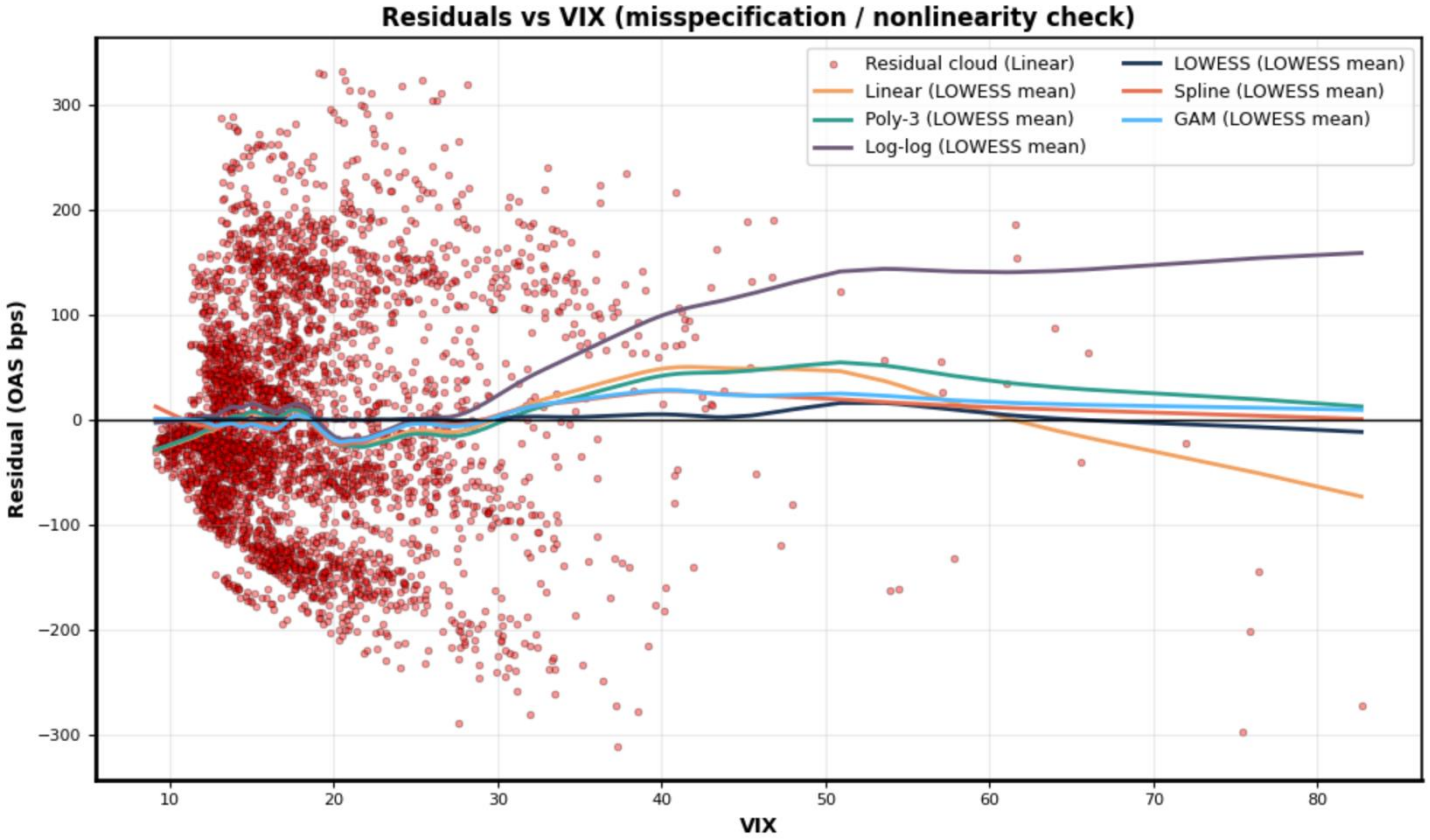
1. **Linear (levels)**
2. **Polynomial (cubic)**
3. **Log–log (power law)**
4. **LOWESS** (nonparametric smoother)
5. **Spline Transformer**
6. **GAM with B-Splines**

Multiple Fits – Regime Colouring VIX Volatility (Low, Mid, High)



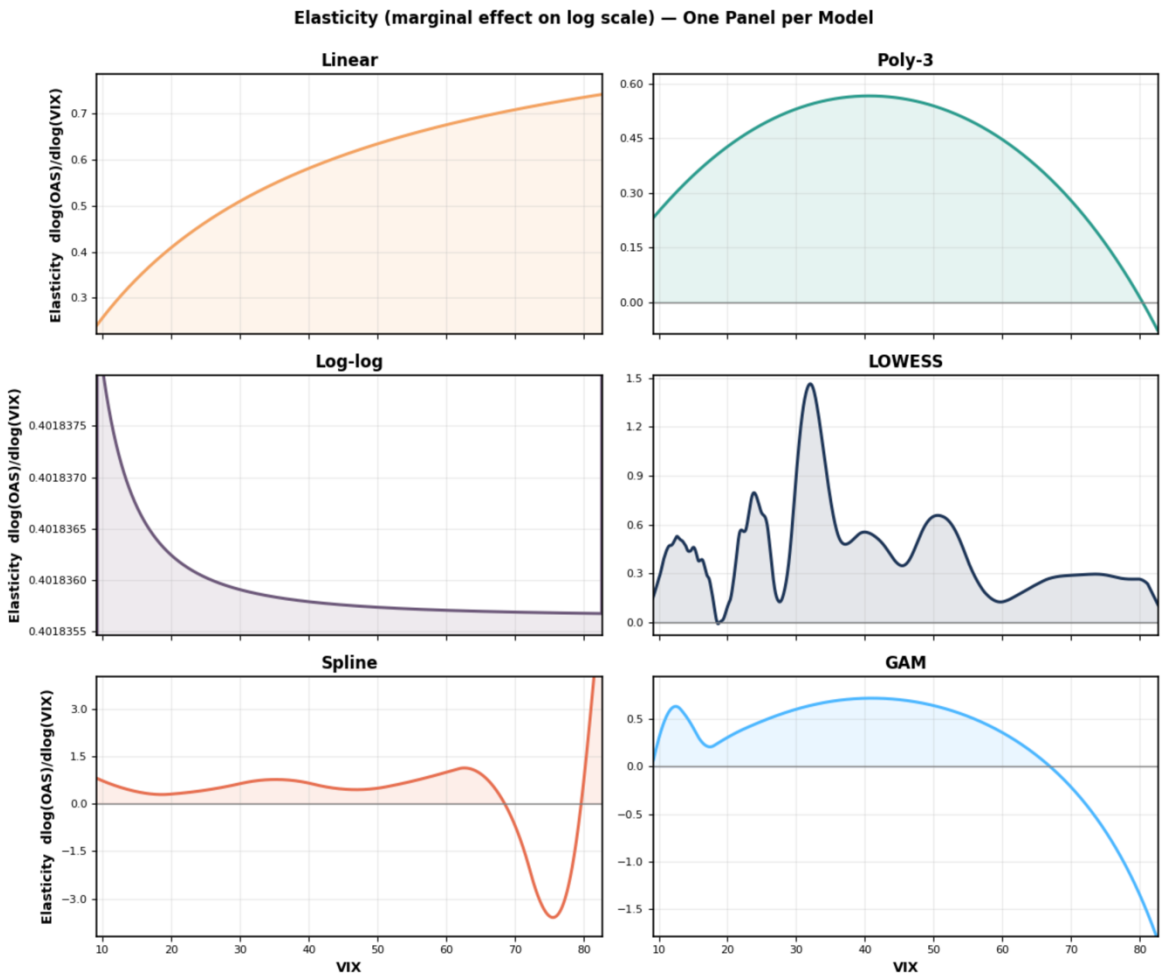
- There in no one stable OAS = f(VIX) relationship, high VIX dispersion widens significantly: **state dependent volatility**
- **VIX** is a **good** indicator of **market stress**, but in **high-stress** periods **OAS becomes much harder to predict** from **VIX alone**
- Variety of models basically indicate: **monotone up (keeps rising** or **flattens)**, **nonlinear**, **tail uncertainty**
- **Conclusion: Regime-conditional model or similar…**

Residuals - Strong Heteroskedasticity and Structure



- Residual dispersion is **very wide** at low-to-mid VIX and remains wide into high VIX
- For conditional mean $\mathrm{E}[OAS \mid VIX]$, the **average residual** should be ~0 at every VIX level
- **Log–log (purple) is far above zero at moderate/high VIX**
- **Mean vs Variance:** models reduce mean-bias but the residuals shows large, regime-dependent variance
- **Regime dependence is real**
- Strong heteroskedasticity in risk-off regimes

Elasticity – Sentitivity Measure of Responce Between Target and Driver variables as a %–age



- **Two notions of sensitivity across regimes**
  - Marginal Effect (marginal effect in levels)
  - Absolute Sensitivity

$$ME(x) \approx \frac{\Delta OAS}{\Delta VIX}$$

  - **Elasticity (log-scale marginal effect)**
  - Normalized sensitivity

$$E(x) \approx \frac{\%\Delta OAS}{\%\Delta VIX}$$

- **Linear - Elasticity rises with VIX but at a decreasing rate(concave – bps/%)**
- **Poly-3 – Elasticity increases into mid VIX then declines (high VIX)**
- Log-Log : Elasticity collapses to the slope coefficient $\beta = here\ 0.4$
  - Does not represent regime variation
  - Slight **curvature** is numerical noise due to **finite-difference derivatives**
- Lowess : Elasticity varies substantially with VIX – State dependence
- Spline : becomes unstable **unstable in the far-right tail**
- GAM : smooth mid-range behaviour – sparse tail

- Use the **log–log elasticity (~0.4)** as the reliable headline figure
- **LOWESS/GAM/Spline -** signs that the VIX–OAS linkage is **regime-dependent** i.e. sensitivity changes across market states
- Focus interpretation on the **mid-range structure** (where data are dense)

Summary – VIX vs. OAS

- **Tested for nonlinearity:** the **relationship is not well-described by a single linear mapping** across the full range
- Nonlinearity manifests as regime-dependent sensitivity**:** the **slope/elasticity changes** between Low Vol, transition, and stress states
- **Different univariate dynamics create nonlinear joint behaviour: fast, mean-reverting** moves in one variable vs **slow, persistent adjustment in the other** produce lags and path dependence
- **Flexible nonlinear models (LOWESS/spline/GAM) reveal curvature and transition regimes**—useful for diagnosing where the relationship bends or steepens
- **Flexible fits are unstable in the tails** because **extreme observations are sparse**; tail curvature can be an artefact without constraints
- **Parsimonious baseline (log–log in this case) gives a stable headline sensitivity**, while regime diagnostics flag when nonlinearity causes deviations

Learnings – Take Away

- **ML can learn nonlinearity implicitly**, but the **learned relationship is often opaque** (shape, thresholds, changes = **unknown**)
- **Explicit nonlinearity analysis makes sensitivity visible** (marginal effects / elasticity) and shows **how it shifts by regime**
- Can exhibit **lags and path dependence** that ML may average away or misattribute
- **Highlights tail fragility** (sparse extremes) reduce the risk of confidence but possible wrong behaviour in stress
- **Such analysis can provide a baseline + diagnostics to validate ML**: stability across regimes, stress performance, and probability calibration